Hugo Marquerie 22/01/2026

Ejer schwarz pick extremales

Ejercicio 1. Sean $\alpha, \beta \in \mathbb{D}$, definimos $\mathcal{A}_{\alpha,\beta} := \{f \in \mathcal{H}(\mathbb{D}) : f(\alpha) = \beta\}$. Demuestra que

(a) $\mathcal{A}_{\alpha,\beta} \neq \varnothing$.

(b) $\displaystyle\sup_{f\in\mathcal{A}_{\alpha,\beta}} |f'(\alpha)| = \frac{1-|\beta|^2}{1-|\alpha|^2}$.

(c) Si $f \in \mathcal{A}_{\alpha,\beta}$ alcanza el supremo, entonces $f = \tau_\beta \circ \rho_\gamma \circ \tau_\alpha$ con $\gamma \in \mathbb{T}$ libre.

***Solución*:**

(a) Basta con tomar $f = \tau_\beta \circ \tau_\alpha$, que es holomorfa y cumple que $f(\alpha) = \tau_\beta(0) = \beta$.

(b) Sea $f \in \mathcal{A}_{\alpha,\beta}$. Por el Teorema de Schwarz-Pick, tenemos que

$$|f'(\alpha)| \leqslant \frac{1-|f(\alpha)|^2}{1-|\alpha|^2} = \frac{1-|\beta|^2}{1-|\alpha|^2}.$$

Además, para $f = \tau_\beta \circ \tau_\alpha$, se satisface la igualdad:

$$\left|(\tau_\beta \circ \tau_\alpha)'(\alpha)\right| = \left|\tau_\beta'(0)\right| \cdot |\tau_\alpha'(\alpha)| \overset{\texttt{lem-involucion-disco-unidad-derivada/1}}{\overset{\downarrow}{=}} \left(|\beta|^2 - 1\right) \cdot \frac{1}{|\alpha|^2-1} = \frac{1-|\beta|^2}{1-|\alpha|^2}.$$

Luego $\displaystyle\sup_{f\in\mathcal{A}_{\alpha,\beta}} |f'(\alpha)| = \frac{1-|\beta|^2}{1-|\alpha|^2}$.

(c) Si $f \in \mathcal{A}_{\alpha,\beta}$ alcanza el supremo, entonces hay igualdad en la segunda desigualdad del teorema de Schwarz-Pick, luego $f \in \operatorname{Aut}(\mathbb{D})$. Definimos $g := \tau_\beta \circ f \circ \tau_\alpha$. Entonces, $g \in \operatorname{Aut}(\mathbb{D})$ y $g(0) = \tau_\beta(f(\alpha)) = \tau_\beta(\beta) = 0$. Por la `obs-aut-disco-unidad-fija-origen-imp-rotacio` $\exists \gamma \in \mathbb{T} : g = \rho_\gamma$. Por tanto, al despejar f, llegamos a $f = \tau_\beta \circ \rho_\gamma \circ \tau_\alpha$. ∎

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.